

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10: SUB-STRAND 2.3 PLANT GASEOUS EXCHANGE AND RESPIRATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation. Use everything you learned in Lessons 1 to 12 to answer the driving question: How do plants "breathe," and what happens inside their cells when they release the energy stored in food? Anchor your answers in the sukuma wiki (kale) bubble phenomenon from Mwalimu Njoroge's class in Nairobi: freshly cut leaves in warm water (about 42°C) gave streams of bubbles from specific spots, while leaves in cold water (about 15°C) gave far fewer. Write in full sentences in the space under each prompt. Use scientific words correctly, give evidence from your own observations and investigations, and explain WHY, not only WHAT. You may draw and label diagrams where a prompt asks. Do not leave out the link between the cell level and the whole-leaf level.

Part 1: Where do the bubbles come from?

In Mwalimu Njoroge's class, bubbles streamed from specific spots on the submerged sukuma wiki leaf, not from the whole surface. (a) Name the structures on the leaf that these bubbles come from, and describe how they are built and where they are found on the leaf. (b) Describe what makes each of these structures change between letting gas through and not letting gas through, using one of the two theories you studied about how this happens. (c) Name the other gas-exchange sites in plants, and say where each is found.

Write your answer here:

Part 2: What is happening inside the cells?

(a) Write the word equation or symbol equation for aerobic respiration and say where in the cell it happens. (b) Name the three stages of aerobic respiration and the part of the cell where each takes place. (c) Compare aerobic and anaerobic respiration in yeast: oxygen needed, products, and ATP yield per glucose. (d) Describe one Kenyan example where anaerobic respiration is useful.

Write your answer here:

Part 3: Why did warm water make more bubbles?

Compare the warm water (about 42°C) and cold water (about 15°C) results. (a) Explain, using enzymes, why temperature changed the number of bubbles. (b) Name the two gases that a submerged leaf could release, the process that makes each, and what each process needs. Then explain what you can and cannot be sure of about the gas in the bubbles you saw, given the tests you did. (c) Describe the evidence from your yeast investigations that supports your temperature explanation, and say which investigation tested temperature and which did not.

Write your answer here:

Part 4: Plants as a linked system

(a) Explain how photosynthesis, gaseous exchange and respiration are linked in a sukuma wiki plant. Say what each process supplies to the others. (b) Answer the misconception: 'Plants only photosynthesise; they do not respire.' (c) Choose one adaptation for gaseous exchange in a xerophyte or a hydrophyte from Kenya and explain how it helps the plant.

Write your answer here:

Part 5: The final explanation

Write a scientific explanation of the anchoring phenomenon for a classmate who missed the unit. Your answer must explain (a) where the bubbles come from and why only from specific spots, (b) what the gases are and what makes them, (c) why warm water gave more bubbles than cold water, and (d) whether plants 'breathe' like animals and what this means at cell level. Finish by connecting the cell level to the whole-leaf level.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Structures and mechanism of gas exchange	Names stomata, guard cells and the other sites correctly and explains opening and closing using osmosis, turgor and K ⁺ or starch-sugar, linked to the bubble spots.	Names stomata and most sites and gives a basic account of guard cell opening, with small gaps in the mechanism.	Names few sites or says bubbles come from the whole leaf; no mechanism for how pores open.
Cellular respiration content	Gives correct equations, location of each stage, ATP yields (36 to 38 versus 2) and compares aerobic and anaerobic products accurately.	Gives correct equations and some locations or ATP figures, with minor errors.	Confuses the products or conditions of aerobic and anaerobic respiration, or omits equations.
Use of evidence and enzyme reasoning for temperature	Uses data from the sukuma wiki and yeast investigations, matches each investigation to what it actually tested, is honest about what was not tested, and explains the temperature effect through enzyme kinetic energy and collisions.	States that warmth speeds up reactions and cites some evidence, with limited reasoning.	States the temperature effect without evidence or reasoning.
Linking photosynthesis, respiration and gas exchange	Clearly shows the two-way flow of CO ₂ , water, O ₂ and glucose, corrects the misconception, and links the cell level to the whole leaf.	Describes the link between the processes, with the cell-to-leaf connection only partly explained.	Treats processes as separate or keeps the misconception that plants do not respire.
Scientific communication and Kenyan context	Writes a clear, well-organised, accurate narrative using correct terms and Kenyan examples.	Writes clearly with mostly correct terms and some local examples.	Writing is unclear or disorganised, with incorrect terms and no examples.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.